$T \propto$ rotor field strength $\times$ stator field strength $\times \sin\delta$

field winding → stator → DC always → establishes & maintains magnetic fields

armature winding → rotor → AC → carries load current / induced emf

field winding → less current

commutator → DC/AC converter → DC/AC inverter

$$E_{coil} = \frac{\phi p \omega_m}{2\pi}$$ $\phi \rightarrow$ flux per pole, $p \rightarrow$ no of poles $\quad \phi \propto I_f$

$$E_g = K_e \phi \omega_m$$

K_e → emf const → controlled by field current

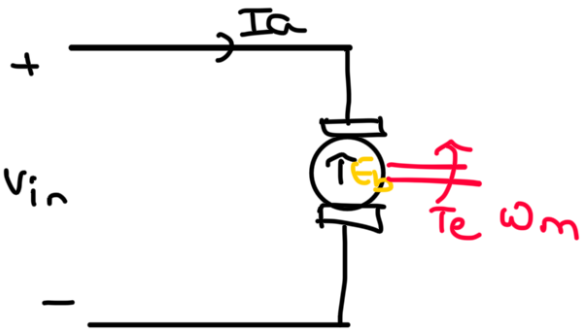
$$T_e = \left(\frac{pz}{2\pi A}\right) \phi I_a$$ → total armature current

$k_t \rightarrow$ torque const $= K_t I_a I_f$

$$V_{in} = E_b + I_a R_a \qquad E_b = K_e \phi \omega$$

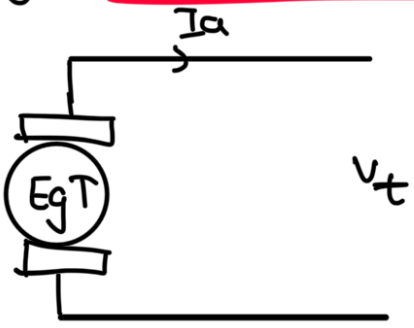
$E_b \rightarrow$ back emf

motor → draws current



$V_{in} = E_b + I_a R_a +$ total brushdrop

generator → supplies current



$E_g = V_t + I_a R_a +$ total brushdrop

Rated power: power it can deliver when fully loaded

(power consumed > Rated power)

Armature voltage control to achieve speeds below

rated speeds

$$\omega_m = \frac{V}{K_e\phi} - \frac{R_a I_a}{K_e\phi} \qquad \omega_m = \underbrace{\frac{V}{K_e\phi}}_{\omega_{mo}} - \underbrace{\frac{R_a T_e}{(K_e\phi)^2}}_{\text{load dependent}}$$

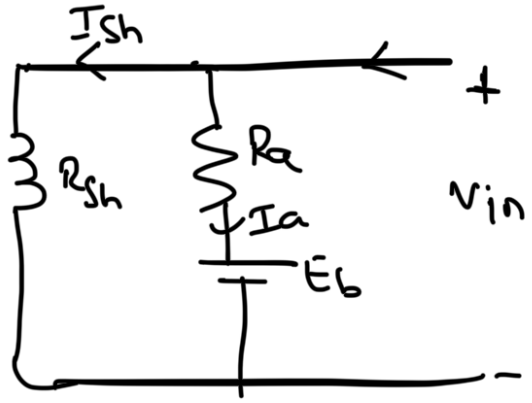
Forward motoring
$\omega_m, T_e > 0$ $P > 0$

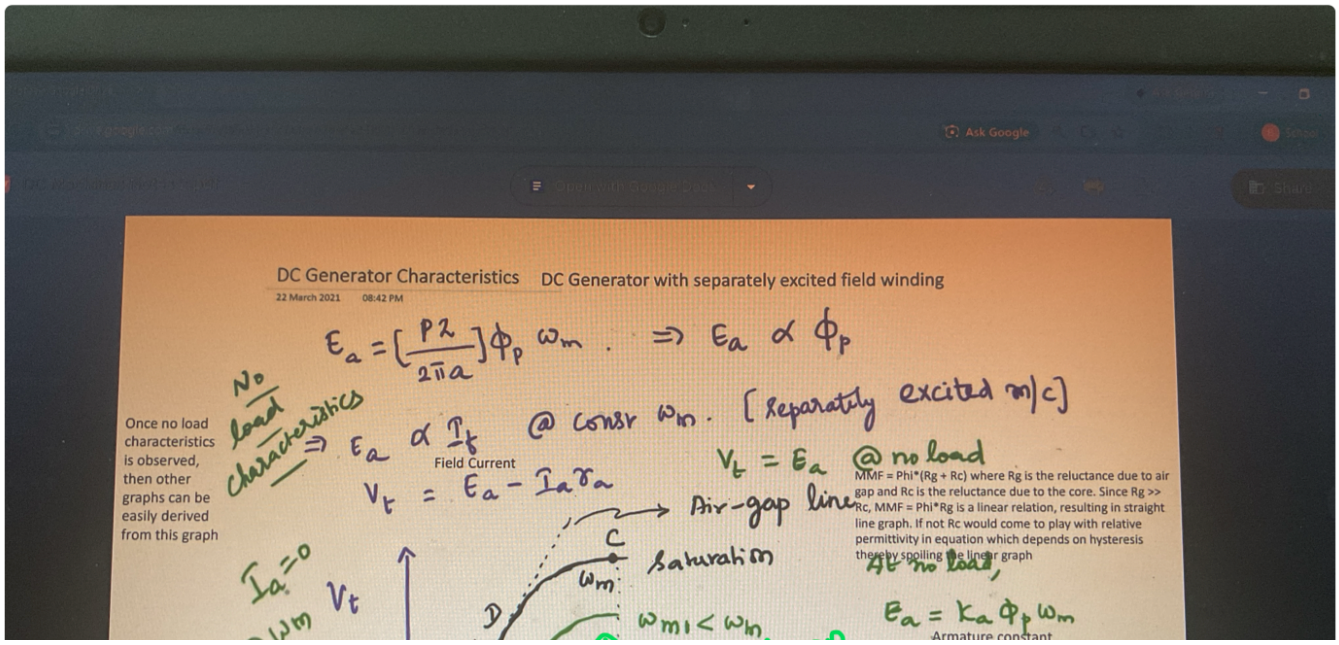
Forward Breaking
$\omega_m > 0$ $T_e < 0$ $P < 0$

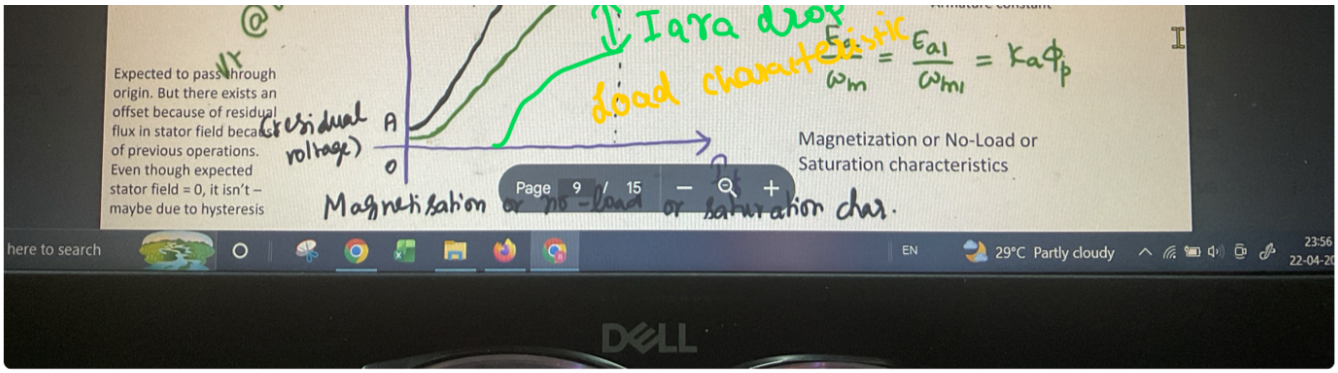
Reverse motoring
$\omega_m < 0$ $T_e < 0$ $P > 0$

Reverse breaking
$\omega_m < 0$ $T_e > 0$ $P < 0$

$$\underbrace{VI_a}_{P_{in}} = \underbrace{E_b I_a}_{P_{out}} + \underbrace{R_a I_a^2}_{\text{armature loss}}$$






Expected to pass through origin. But there exists an offset because of residual flux in stator field because of previous operations. Even though expected stator field = 0, it isn't – maybe due to hysteresis
(residual voltage)
A
0
Iara drop
load characteristic
$\frac{E_a}{\omega_m} = \frac{E_{a1}}{\omega_{m1}} = K_a \phi_p$
Magnetization or No-Load or Saturation characteristics
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Magnetisation or No-load or saturation char.